A few notes...

- **HW6 should be graded and published in your Canvas grade center by tonight; HW8 is due 11:59pm Tues Apr 14**

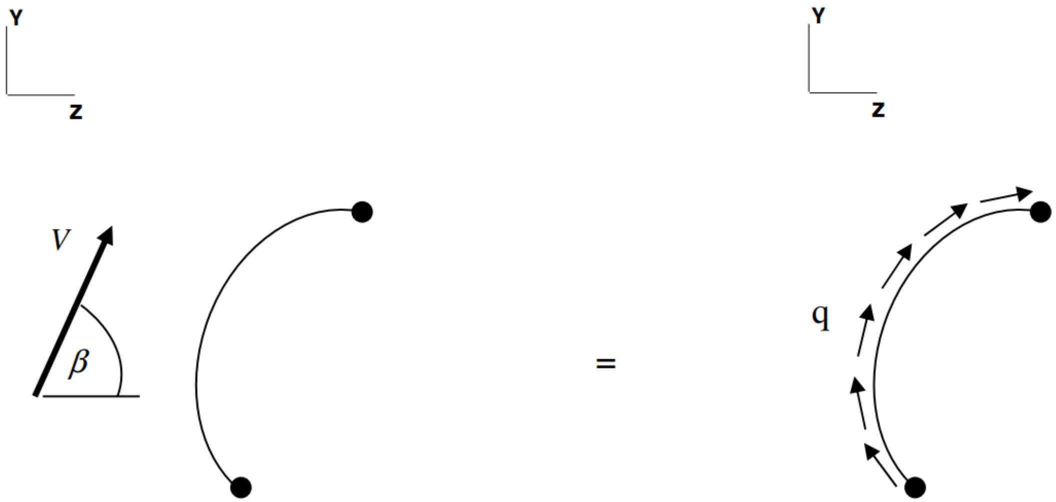
- **I decided hold the next exam on Mon Apr 20, *NOT* Fri Apr 17 as originally planned**

- **Here's roughly how our remaining class periods will go:**
  - 4/10 (today): Cover more Chapter 5 material (Sec 5.3)
  - 4/13: Cover remaining Chapter 5 material and go over HW6 solutions
  - 4/15: Cover some Chapter 6 (or 8) material
  - 4/17: Go over HW8 solutions and TEST PREP: Go over review problems, answer questions, etc
  - 4/20: 2^{nd} Exam
  - 4/22: Go over and prep for the Final (likely take-home)

5.3: Idealized Cross-Section Beams Under Transverse Loading
Curved Webs / Closed Cross-Sections

- **Up to now we've been analyzing idealized beams with open cross sections consisting of straight webs between the stringers; let us now consider beams with closed cross sections consisting of curved webs between the stringers**

- **The figure below shows equivalent interior faces for an idealized beam under transverse loading**
  - Two effective stringers connected by a single curved web.
  - Left face shows interior shear force V (acting at the shear center) due to the transverse loads
  - Right face shows the resulting shear flow distribution

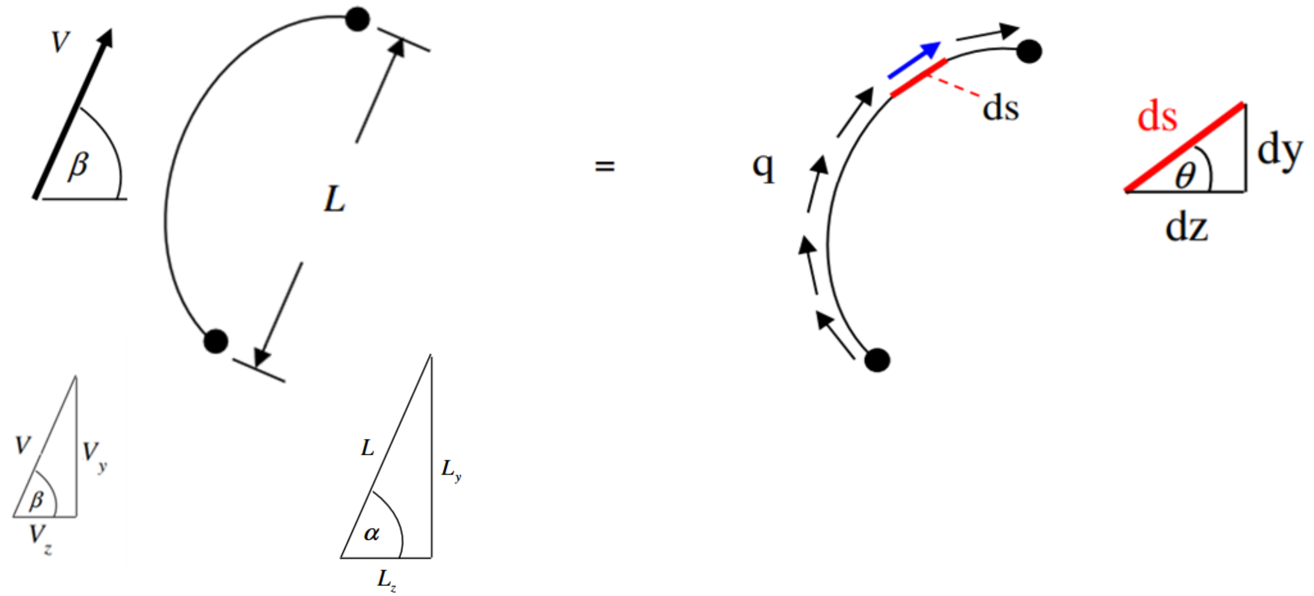


5.3: Idealized Cross-Section Beams Under Transverse Loading
Curved Webs / Closed Cross-Sections

- **Force equivalence dictates that if we integrate (i.e. sum up) the resulting shear flow distribution on the right, it must equal the shear force *V***

- **Since this is an idealized cross-section, the magnitude of *q* is constant along the web, however the direction of *q* is now changing since the webbing is curved**

- **In order to solve the relationship between *q* and *V*, let's add a bit more to this diagram:**



5.3: Idealized Cross-Section Beams Under Transverse Loading
Curved Webs / Closed Cross-Sections

- **Summing forces in the *y* direction yields:**

$$\sum\left(F_y\right)_{\text{Left Figure}} = \sum\left(F_y\right)_{\text{Right Figure}}$$

$$\Rightarrow \quad V_y = \int_{\substack{\text{Curved}\\ \text{Length}}} (q\,ds)\sin\theta$$

$$\Rightarrow \quad V_y = \int_{\substack{\text{Curved}\\ \text{Length}}} q\,\overbrace{ds\sin\theta}^{dy}$$

$$\Rightarrow \quad V_y = \int_0^{L_y} q\,dy$$

$$\Rightarrow \quad \boxed{V_y = q L_y}$$

5.3: Idealized Cross-Section Beams Under Transverse Loading
Curved Webs / Closed Cross-Sections

- **Summing forces in the *z* direction yields:**

$$\sum \left(F_z\right)_{\text{Left Figure}} = \sum \left(F_z\right)_{\text{Right Figure}}$$

$$\Rightarrow \quad V_z = \int_{\substack{\text{Curved}\\\text{Length}}} \left(q\,ds\right)\cos\theta$$

$$\Rightarrow \quad V_z = \int_{\substack{\text{Curved}\\\text{Length}}} q\,\overbrace{ds\cos\theta}^{dz}$$

$$\Rightarrow \quad V_z = \int_0^{L_z} q\,dz$$

$$\Rightarrow \quad \boxed{V_z = qL_z}$$

5.3: Idealized Cross-Section Beams Under Transverse Loading
Curved Webs / Closed Cross-Sections

- **In addition to these expressions for V_y and V_z, we can also derive expressions the magnitude and direction of V:**

$$V = \sqrt{(V_y)^2 + (V_z)^2}$$

$$= \sqrt{(qL_y)^2 + (qL_z)^2}$$

$$= q\sqrt{(L_y)^2 + (L_z)^2}$$

$$\boxed{V = qL}$$

$$\tan\beta = \frac{V_y}{V_z}$$

$$= \frac{qL_y}{qL_z}$$

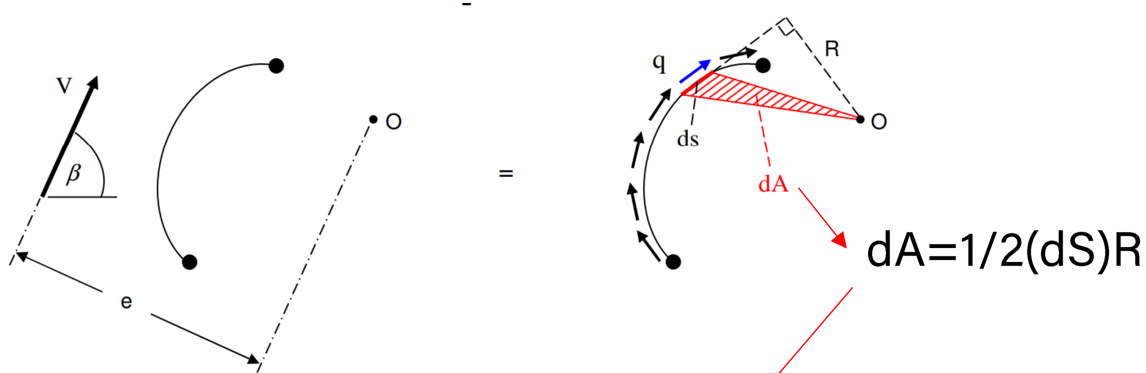
$$= \tan\alpha$$

$$\alpha = \beta$$

$$\boxed{V \parallel L}$$

- **The implications of this are that the shear force equivalent to the q distribution along a curved web between two stringers is the same as it would be along a straight web between the two stringers**

- **What if we desired to find the location where *V* must act so as to incur no torsion? We can do this by performing moment equivalence about an arbitrarily defined point O:**



$$\circlearrowleft + \quad \sum\left(M_0\right)_{\text{Left Figure}} = \sum\left(M_0\right)_{\text{Right Figure}}$$

$$Ve = \int_{\substack{\text{Curved}\\\text{Length}}} R(q\,ds)$$

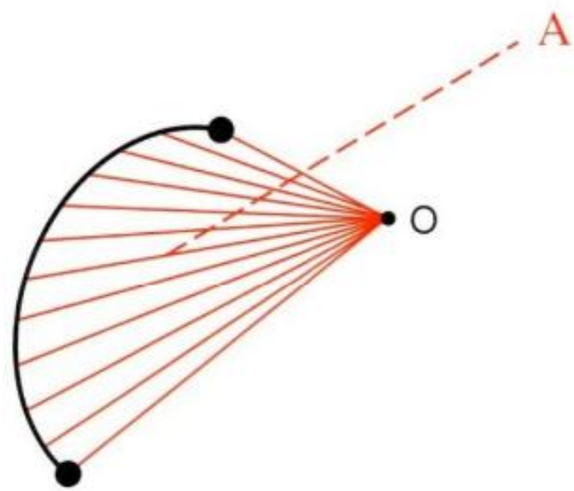
$$Ve = \int_{\substack{\text{Curved}\\\text{Length}}} q\,\overbrace{Rds}^{2dA}$$

$$Ve = 2q\int_{Area} dA$$

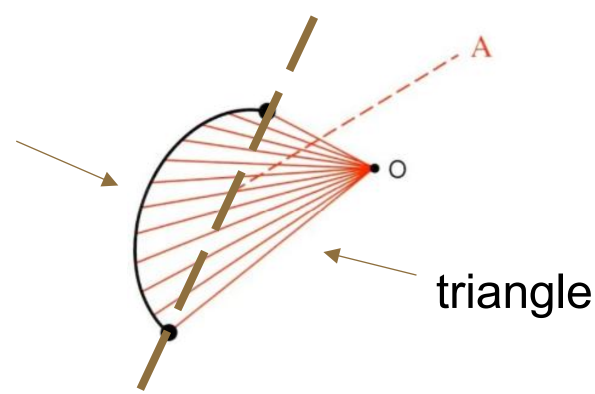
$$\boxed{Ve = 2Aq}$$

5.3: Idealized Cross-Section Beams Under Transverse Loading
Curved Webs / Closed Cross-Sections

- **Here A is the area swept along the web for all of the q values, using Point O as the pivot point; i.e. it is the area shown below:**

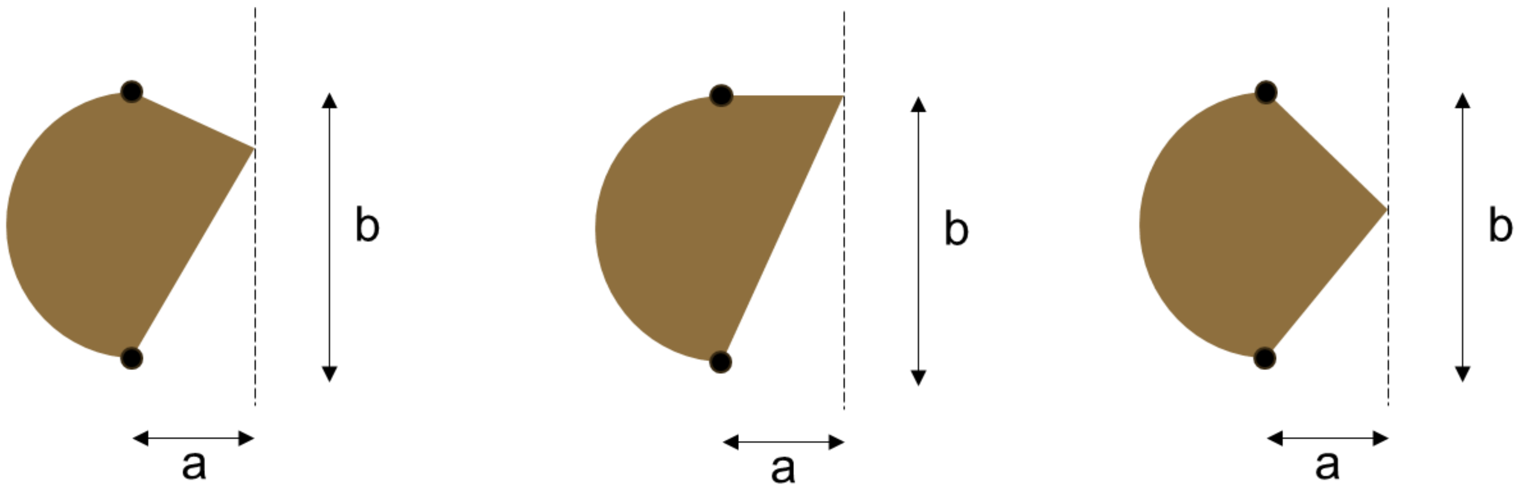


- **Note that if the curved web forms an arc of a circle, this area partitions nicely into the fractional area of that circle plus the area of a triangle**



5.3: Idealized Cross-Section Beams Under Transverse Loading
Curved Webs / Closed Cross-Sections

- **Note also that the swept area (and therefore moment) is the same at any point along a line drawn parallel to the line between the 2 stringers:**



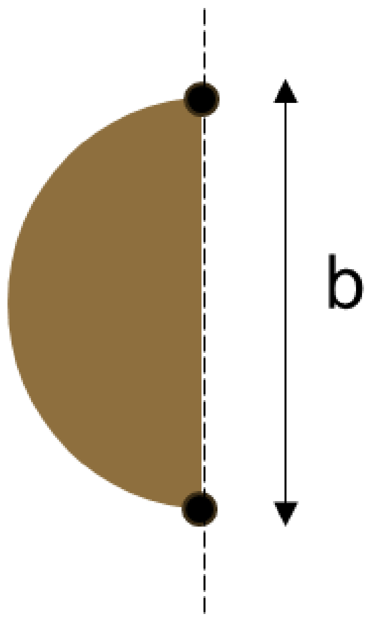
$A_{swept} = A_{circ} + (½)ab$ $\quad$ $A_{swept} = A_{circ} + (½)ab$ $\quad$ $A_{swept} = A_{circ} + (½)ab$

- **Where A_{circ} represents the area of the sector of the circle and $(½)ab$ is the triangular area**

- **If the sector happens to be a semi-circle, then $A_{circ} = (½)\pi(b/2)^2$**

5.3: Idealized Cross-Section Beams Under Transverse Loading
Curved Webs / Closed Cross-Sections

- **And it's a bit more difficult to visualize, but the swept area is also the same at any point along the line between the 2 stringers:**



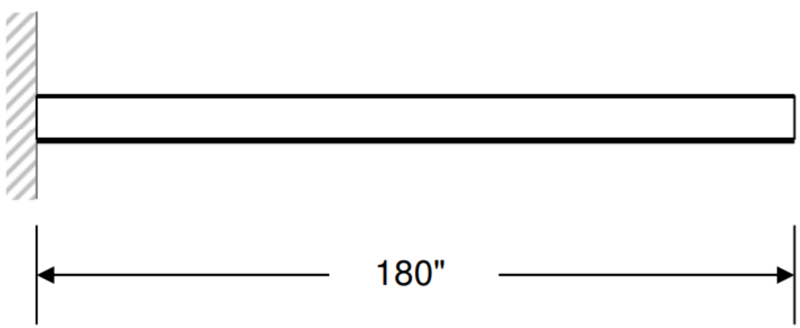
$$A_{swept} = A_{circ}$$

- **Again, if the sector happens to be a semi-circle, then** $A_{circ} = (½)\pi(b/2)^2$

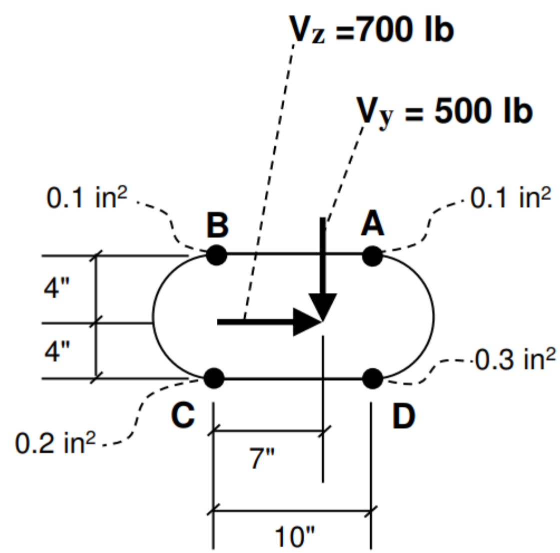
Example 5.3.1

- **For the beam and cross-section shown below, find the shear flow at the wall for each web of the cross-section**

Span View



Interior Cross-Section View
Showing
Shear Forces



(Both Curves are Semicircles)

Web Thicknesses = 0.040”

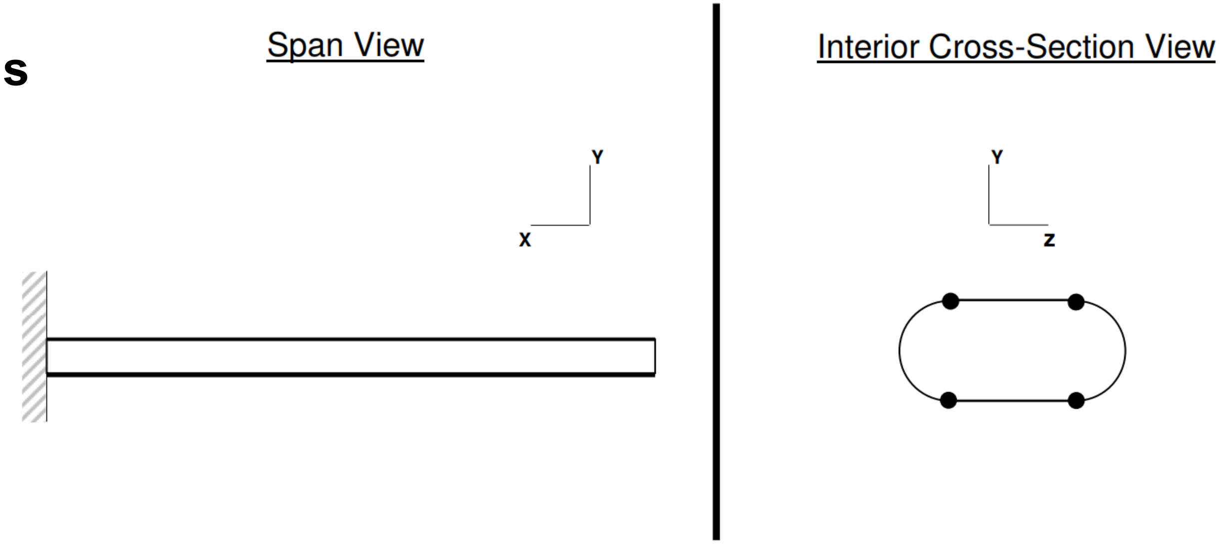
Example 5.3.1

- **SIDE NOTE: The textbook doesn't mention it, but in this example we are to assume that V_y and V_z are acting at the shear center**
- **In the previous section we learned how to calculate the shear center for open cross-sections; for closed cross-sections, a different technique is required (not covered in the textbook)**
- **The GSFF is derived assuming forces are acting at the shear center, therefore q is due only to shear force (i.e. no torsion)**
- **In this section we are to make that assumption, therefore we can use the GSFF to fully express q**
- **If there were torsion, we'd have to account for it as well in calculating q**

Example 5.3.1

- **We know the general steps to follow:**
  - Choose axis directions
  - Calculate centroid of cross section and move coordinate frame origin to the centroid
  - Calculate inertias (I_y, I_z, I_{yz})
  - Determine sign and value of shear force components at wall (V_y and V_z)
  - Insert V_y, V_z, I_y, I_z, I_{yz} values into GSFF

- **Choose our usual axis directions like this →**



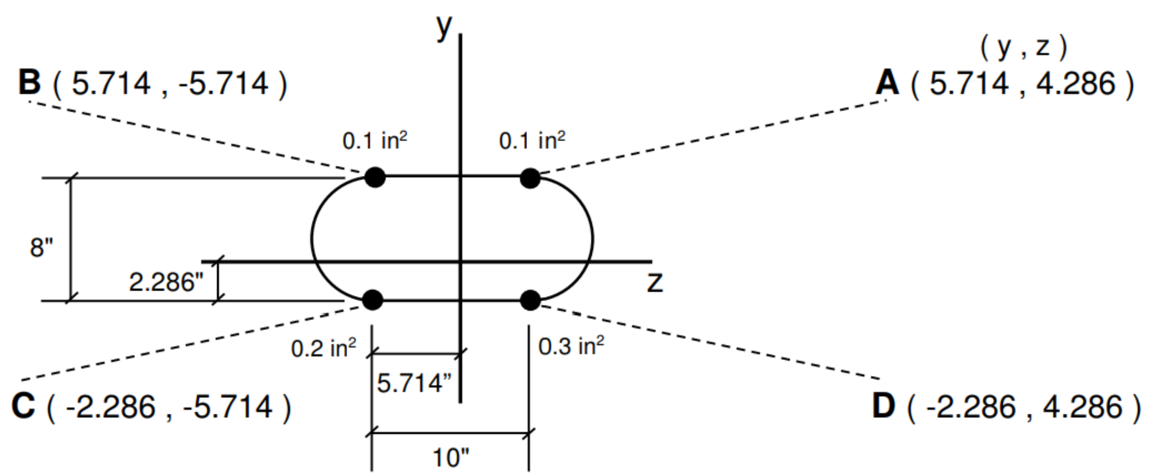
Example 5.3.1

- **Calculate the centroid of the cross section (placing our temporary *y* and *z* axes at point C):**

$$\bar{y} = \frac{2\left(0.1in^2\right)\left(8in\right)}{0.7in^2} = 2.286in$$

$$\bar{z} = \frac{\left(0.1in^2\right)\left(10in\right) + \left(0.3in^2\right)\left(10in\right)}{0.7in^2} = 5.714in$$

- **Then place the *y*-*z* frame at the centroid and find the centroidal coordinates of each stringer:**



Example 5.3.1

- **Calculate inertias (remembering that we only need the "parallel axis" term for each stringer):**

$$I_y = \sum_i A_i z_i^2$$

$$= \left\{ \begin{aligned} &(0.1in^2)(4.286in)^2 + (0.1in^2)(-5.714in)^2 \\ &+(0.2in^2)(-5.714in)^2 + (0.3in^2)(4.286in)^2 \end{aligned} \right\}$$

$$\boxed{I_y = 17.14in^4}$$

$$I_z = \sum_i A_i y_i^2$$

$$= \left\{ \begin{aligned} &(0.1in^2)(5.714in)^2 + (0.1in^2)(5.714in)^2 \\ &+(0.2in^2)(-2.286in)^2 + (0.3in^2)(-2.286in)^2 \end{aligned} \right\}$$

$$\boxed{I_z = 9.143in^4}$$

$$I_{yz} = \sum_i A_i y_i z_i$$

$$= \left\{ \begin{aligned} &(0.1in^2)(5.714in)(4.286in) + (0.1in^2)(5.714in)(-5.714in) \\ &+(0.2in^2)(-2.286in)(-5.714in) + (0.3in^2)(-2.286in)(4.286in) \end{aligned} \right\}$$

$$\boxed{I_{yz} = -1.143in^4}$$

Example 5.3.1

- **Determine sign and value of shear force components at wall:** *V_y = +500 lb*, *V_z = -700 lb*

- **Insert values into GSFF and simplify:**

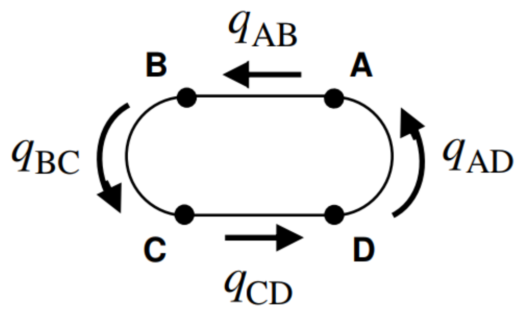
$$q_{out} - q_{in} = \frac{\left(I_y V_y - I_{yz} V_z\right)\overline{y}' A' + \left(-I_{yz} V_y + I_z V_z\right)\overline{z}' A'}{I_y I_z - I_{yz}{}^2}$$

$$q_{out} - q_{in} = \frac{\left\{\begin{array}{l}\left[\left(17.14 in^4\right)\left(+500 lb\right) - \left(-1.143 in^4\right)\left(-700 lb\right)\right]\overline{y}' A' \\ +\left[-\left(-1.143 in^4\right)\left(+500 lb\right) + \left(9.143 in^4\right)\left(-700 lb\right)\right]\overline{z}' A'\end{array}\right\}}{\left(17.14 in^4\right)\left(9.143 in^4\right) - \left(-1.143 in^4\right)^2}$$

$$\boxed{q_{out} - q_{in} = \left[\left(50.00\right)\overline{y}' - \left(37.51\right)\overline{z}'\right] A'}$$

Example 5.3.1

- **Now let's label each of our shear flows on the cross-section and their direction:**



- **So at Stringer A we have:**

$$q_{AB} - q_{AD} = \left[(50.00)(5.714) - (37.51)(4.286)\right]\left(0.1in^2\right)$$

$$\Rightarrow \quad \boxed{q_{AB} = q_{AD} + 12.49}$$

- **At Stringer B we have:**

$$q_{BC} - q_{AB} = \left[(50.00)(5.714) - (37.51)(-5.714)\right]\left(0.1in^2\right)$$

$$\Rightarrow \quad q_{BC} = \overbrace{\left(q_{AD} + 12.49\right)}^{q_{AB}} + 50.00$$

$$\Rightarrow \quad \boxed{q_{BC} = q_{AD} + 62.49}$$

Example 5.3.1

- **At Stringer D we have:**

$$q_{AD} - q_{CD} = \left[(50.00)(-2.286) - (37.51)(4.286)\right]\left(0.3in^2\right)$$

$$\Rightarrow \qquad \boxed{q_{CD} = q_{AD} + 82.52}$$

- **At Stringer C we have:**

$$q_{CD} - q_{BC} = \left[(50.00)(-2.286) - (37.51)(-5.714)\right]\left(0.2in^2\right)$$

$$\Rightarrow \qquad q_{CD} = \overbrace{\left(q_{AD} + 62.49\right)}^{q_{BC}} + 20.01$$

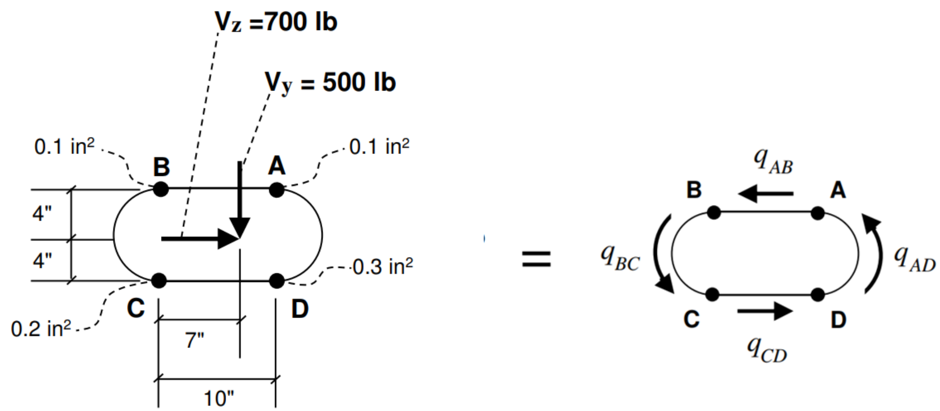
$$\Rightarrow \qquad \boxed{q_{CD} = q_{AD} + 82.50}$$

- **But accounting for roundoff error, this is the same equation as the one above!**

- **We need a 4th equation to solve these shear flows…**

Example 5.3.1

- **The 4th equation can be obtained from moment equivalence between the two figures below:**



- **Choosing to equate moments about point C, we have:**

$$\circlearrowleft + \quad \left(\sum M_C\right)_{\text{Left Figure}} = \left(\sum M_C\right)_{\text{Right Figure}}$$

$$-500\text{lb}*7\text{in} - 700\text{lb}*4\text{in} = (q_{AB}\ \text{lb/in})*10\text{in}*8\text{in} + 2*A_{swept,qBC}*(q_{BC}\ \text{lb/in}) + 2*A_{swept,qAD}*(q_{AD}\ \text{lb/in})$$

Aerospace Engineering Department
College of Engineering
Daytona Beach Campus

Example 5.3.1

- **From slide 10, we see that** $A_{swept,qBC} = (½)\pi(4)^2$ **=** $8\pi\ in^2$

- **From slide 9, we see that** $A_{swept,qAD} = (½)*10*8 + (½)\pi(4)^2 = (40 + 8\pi)in^2$

- **From slide 16, we see that** $q_{AB} = q_{AD} + 12.49$ **and** $q_{BC} = q_{AD} + 62.49$

- **Inserting these into the equation on the previous slide and solving for** q_{AD} **yields** $q_{AD} = -40.07\ lb/in$

- **Inserting this into the equations on Slides 16-17 yields →**

$$q_{AB} = -27.58\ \frac{lb}{in}$$

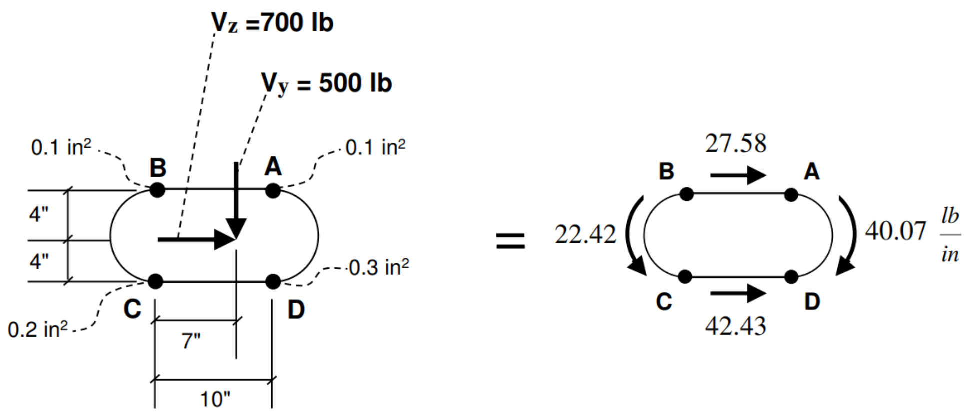
$$q_{BC} = +22.42\ \frac{lb}{in}$$

$$q_{CD} = +42.43\ \frac{lb}{in}$$

*******DONE!!!!*******

Example 5.3.1

- We can run a check on the answers by using force equivalence in both the *y* and *z* directions for the two figures below, remembering that $V = qL$:



$$\left(\sum F_y\right)_{\text{Left Figure}} = \left(\sum F_y\right)_{\text{Right Figure}}$$

$$500\ lb = \left(22.42\ \frac{lb}{in}\right)(8") + \left(40.07\ \frac{lb}{in}\right)(8")$$

$$= 499.92\ lb$$

Checks ✓

$$\left(\sum F_z\right)_{\text{Left Figure}} = \left(\sum F_z\right)_{\text{Right Figure}}$$

$$700\ lb = \left(27.58\ \frac{lb}{in}\right)(10") + \left(42.43\ \frac{lb}{in}\right)(10")$$

$$= 700.1\ lb$$

Checks ✓